Multimodal Technologies and Interaction

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Systematic Review

Extended Reality Applications for CNC Machine Training: A Systematic Review

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Abstract: Extended reality (XR) as an immersive technology has gained significant interest in the industry for training and maintenance tasks. It offers an interactive, three-dimensional environment that can boost users' efficiency and safety in various sectors. The present systematic review provides information based on a Scopus database search for research articles from 2011 to 2024 to expose 19 selected studies related to XR developments and approaches. The purpose is to grasp the state of the art, focusing on user training in goals or tasks that involve computer numerical control (CNC) machines. The study revealed approaches that broadly employed XR devices to execute diverse operations for virtual CNC machines, offering enhanced safety and skills acquisition, lessening the use of physical machines that impact energy consumption or the time invested by an expert worker to teach an operation task. The articles highlight the advantages of XR training versus traditional training in CNC machines, revealing an opportunity to enhance learning aligned to the industry 4.0 (I4.0) paradigm. Virtual reality (VR) and augmented reality (AR) applications are the most used and are mainly centered on a single-user environment. In addition, a VR approach is built as a proof of concept for learning CNC machine operations, considering the key features identified.

Keywords: virtual reality; disruptive teaching; digital transformation; computer numerical control

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1. Introduction

Industry 4.0 (I4.0) drives using disruptive technology, where innovation completely revolutionizes the work process [1]. I4.0 refers to implementing technologies related to pervasive digitalization in smart manufacturing processes such as virtual reality (VR) and the Internet of Things [2]. For instance, VR replicates a company's facility to support decision making in manufacturing systems and provides a deep understanding of a simulation [3]. This digital transformation is challenging for industry and academic sectors because traditional and passive learning methods are becoming obsolete [4].

The adoption of I4.0 core technologies such as VR, augmented reality (AR), and mixed reality (MR) for training tasks have unique characteristics and promise to open new dimensions in the user experience [5]. These technology solutions modify how the user interacts and perceives information from the environment [6], offering training customization [7] or enhancing worker efficiency [8]. Regarding immersion, VR has reached its full potential to immerse users in exploring virtual worlds through head-mounted displays (HMD) and great advancements in haptic devices [9].

parameters [93]. On the other hand, operators of CNC machines need different, specialized, and broad skills, while expert operators are scarce. Henceforth, training novel users implies high costs [91]. In I4.0, cognitive DT combines human tacit knowledge with DT models with the ability to monitor the status of the corresponding manufacturing elements, deal with unforeseen situations, and hence support decisions [94]. In this sense, capturing the expert knowledge of CNC operators is essential for expertise-sharing paradigms to train new operators by achieving digital identities not only with technical knowledge and skills but also with ways of thinking, ways of acting, or reasoning [95], and transferring them to the VE or metaverse.

The meta-operator emerges from the fusion of metaverse and industry 5.0 to address the worker's knowledge and skills, using the latest XR technologies [96]. The operator requires the support of haptic devices to replicate the realism of virtual elements and direct hand–object manipulation through tactile and kinesthetic feedback. Therefore, the use and research of new materials (e.g., nanocomposites), multimodal sensory technologies [97], the combination of driving mechanisms (e.g., electro-adhesion with ultrasonic signals), and the integration of different haptic cues are essential [98]. The goal is to transmit sensations to the mechanoreceptors of human skin [99], enabling the sense of touch to deliver tactile experiences and skills transfer [100]. Concerning maintenance tasks, the prognosis is verified by an AI avatar to aid in repair planning and training for maintenance technicians. Moreover, a haptic-based robot teleoperation system for blade repair grinding has been developed [101].

6.1. Implications

The present research attempts to provide an overview of the state of the art concerning the XR developments and approaches focused on user training in tasks and operations of CNC machines in academic and industrial domains. The insights presented reflect a snapshot of the current state of the literature, pointing out that the use of XR technologies for training improves spatial and psycho-motor skills [26,102], enables real-time decision-making capability [103], and aids in dealing with the operator's cognitive load due to the complexities of CNC technology [43,91]. In addition, regarding hardware, HMD equipment, mobile devices, and glasses are the most employed in training, and there is a growing research interest in haptic feedback capabilities to enrich visual and auditory user channels that enhance the immersion experience in VEs.

6.2. Limitations

The main limitation of this study is that the authors identified few eligible studies, mainly due to the decision to restrict the search only to a well-known database such as Scopus, as well as the search filters applied (e.g., document type, date range, and language), which could have left out other relevant studies. Further studies should consider conducting in-depth research, opening the date period, and covering other studies from scientific venues such as congresses and conferences, supported by a bibliometric analysis to discover a link among the XR applications, researchers, and human interaction technologies. In light of the review findings, developers, researchers, and policy-makers can consider the actions to take concerning the design and the use of XR technologies. In this vein, this enables the creation of training programs that incorporate multimodal human–computer interaction to enrich user experiences in industrial and academic contexts.

6.3. Future Research

This systematic review depicted XR applications for CNC machine training, showing a panorama to explore advancements in academic and industrial sectors. The design of a proof of concept helps contrast theoretical versus practical issues and the advantages and drawbacks of XR technology. A couple of research directions can be pursued initially to develop an ontology that integrates expert knowledge related to the settings and operation of CNC machines and to create an AI avatar to interact with the trainee in the VE. Second,

the design and construction of low-cost haptic devices with several ad hoc shapes to the objects in virtual scenes enable more immersive and interactive VEs, for instance, working tools according to their physical shape, displaying their main features, and supporting the work on the assembly of a CNC machine.

7. Conclusions

XR technology is pivotal in transforming CNC machine training methodologies, providing immersive and risk-free environments for skills development. The literature review illustrated diverse dimensions of XR applications for CNC machine training, including VR, AR, and MR. The study revealed a landscape ripe with innovation for industrial training. In this sense, one study indicated that HF devices improved user experience and training outcomes, emphasizing the importance of sensory feedback in XR environments [81]. Regarding the hardware and software attributes, HMD equipment, mobile devices, and glasses were the most commonly reported, and the Unity engine was used for XR applications development and Vuforia to extend functionalities. In addition, MR applications stood out for their ability to modify virtual object behavior and enhance realism to immerse the user in the highest learning experience.

On the other hand, most XR applications were deployed in academic settings in classrooms and a few in engineering laboratories, being tested primarily on students and educational personnel such as teachers and laboratory technicians. Developing collaborative XR environments fosters teamwork and knowledge exchange, promising further advancements in skills development in the industrial metaverse. Overall, XR presents a paradigm shift in academic and industrial education, offering enhanced safety and skills acquisition, lessening the use of physical machines that impact energy consumption or the time invested by an expert worker in a training task, all of the above without the need to stop or interrupt a production process. Finally, the proof of concept presented a VR development that offers a tool to train users on typical tasks of the operation of a CNC machine. This approach facilitates envisioning the potential of these kinds of applications.

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References

1. Basile, V.; Tregua, M.; Giacalone, M. A three-level view of readiness models: Statistical and managerial insights on industry 4.0. *Technol. Soc.* **2024**, *77*, 102528. [CrossRef]
2. Cimino, A.; Longo, F.; Mirabelli, G.; Solina, V.; Verteramo, S. An ontology-based, general-purpose and industry 4.0-ready architecture for supporting the smart operator (part ii–virtual reality case). *J. Manuf. Syst.* **2024**, *73*, 52–64. [CrossRef]
3. Huerta-Torruco, V.A.; Hernandez-Uribe, O.; Cárdenas-Robledo, L.A.; Rodríguez-Olivares, N.A. Effectiveness of virtual reality in discrete event simulation models for manufacturing systems. *Comput. Ind. Eng.* **2022**, *168*, 108079. [CrossRef]
4. Azevedo, A.; Almeida, A.H. Grasp the challenge of digital transition in smes—A training course geared towards decision-makers. *Educ. Sci.* **2021**, *11*, 151. [CrossRef]
5. Cárdenas-Robledo, L.A.; Hernández-Uribe, Ó.; Reta, C.; Cantoral-Ceballos, J.A. Extended reality applications in industry 4.0.–A systematic literature review. *Telemat. Inform.* **2022**, *73*, 101863. [CrossRef]
6. He, Z.; Sui, X.; Jin, G.; Cao, L. Progress in virtual reality and augmented reality based on holographic display. *Appl. Opt.* **2019**, *58*, A74–A81. [CrossRef] [PubMed]